Firefighter Bot Proposal

TEJ4MI

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Description

Challenge Statement

Although my firefighter bot may not particularly stand out in physical design and structure, my bot will have a unique software and actual configuration in the maze. My bot will feature a triple layer design with the dimensions 15 cm x 15 cm x 21 cm .I will rely on my creativity and critical problem solving to design a unique bot in terms of actual performance. By analyzing past success and entering this project with a determined mindset, I intend to complete as much as both mazes as possible, as fast as possible. I will be able to test multiple strategies and experiment with different code to figure out which approaches work the most efficiently and effectively.

Line Detection

My bot will use a line detection board to navigate through the maze and recognize different rooms. A superbright LED will be shined onto the floor with a Vishay BPW17N phototransistor to detect the reflected light. Whenever the bot is on a black surface, there will be little to no reflected light, which keeps the voltage above 1.8 volts, resulting in a digital signal of 1. On the other hand, when the bot crosses a white line, the light from the superbright LED will reflect light, which will drop the voltage below 0.8 volts, outputting a digital signal of 0. Whenever the voltage from the pin that is connected to the output of the line detection drops to 0, the bot will detect that a line has been crossed. Using these inputs from the line detection, my bot can keep track of the lines and understand which room it is currently in, aiding my maze navigation.

Wall Detection

My bot will feature two Parallax 28015 distance sensors, with one mounted to the left of the chassis and the other one mounted to the front. With these two sensors, I will be able to apply a wall-hugging algorithm that will serve as the foundation for more complex strategies. my bot will focus on maintaining a distance of 12 centimeters from the wall to the left. If the distance from the left sensor input is greater than 12 centimeters, the bot will temporarily pause the left motor so that the right motor can force the bot to veer slightly leftward, reducing the distance until it reaches 12 centimeters again. Conversely, if the left sensor input reads a distance of less

than 12 centimeters, the bot will pause the right motor, causing the bot to veer rightward instead, increasing the distance until it reaches 12 centimeters again. Whenever the distance is right around 12 centimeters, the bot will proceed forward as usual. Since I am relying on the distance from the left wall, left turns will happen by default, but I still need to apply additional programming to take right turns into consideration. When the front sensor does not detect any walls, the bot can continue to hug the left wall. However when both the left and front sensors are detecting a wall, that means the only possible path is rightward. In this case, there will be a right turn function, turning the left motor on and reversing the polarity on the right motor to make it go backward. This will make the bot spin clockwise, and I will perform tests to determine the exact length of time the bot will spend spinning for consistent turns. This wall-hugging algorithm covers all the cases required for basic maze navigation, providing a solid foundation for me to build upon.

Flame Detection and Board

To handle fire detection and distinguishing, my bot will utilize a QSD123 infrared phototransistor along with a 103k Ω resistor to detect the specific light signatures of a candle flame. Once a flame is detected, the bot will trigger the fire distinguisher board to distinguish the flame. This board is designed to be consistent and reliable, utilizing a TIP120 semiconductor as a switch to activate the distinguishing mechanism. To protect the sensitive logic components from logic spikes caused by the motor or pump, a 1N4001G diode is incorporated into the board for flyback protection, while a 1000 μF capacitor is used to stabilize the power supply to prevent resets during high-draw moments. A 10k Ω resistor is also integrated into the circuit to ensure the transistor switches cleanly and efficiently. By combining this precise detection hardware with a heavy-duty power delivery circuit with filtering, the bot can seamlessly transition from its wall-hugging navigation to an active fire distinguishing mode when a heat source is detected.

Motherboard and Motor Driver

The brains of my bot will reside on a custom motherboard centered around the ATMEGA328P-PU microchip, which processes sensor data and executes navigation logic. To output real-time feedback and debugging information, I will integrate an NHD-0216K1Z-NSW-FBW-L LCD display, with a 100k Ω potentiometer to allow for precise

contrast adjustment for readability. Additionally, a 233k Ω resistor will be utilized within the board to ensure stable signal references for the processor. For actual movement, the bot will carry a pair of high-torque Nema 17 stepper motors secured with strong motor brackets designed specifically for the Nema 17 motors. These motors will be driven by TI SN754410NE motor driver chip; a quadruple high-current, half-H driver, capable of bidirectional control. In order to handle the significant current spikes that are to be expected from the stepper motors, a powerful 10,000 μF capacitor is placed on the power fail to smooth out voltage ripples, while 33k Ω resistors are used to pull the logic pins to their appropriate states, ensuring the motors respond reliably to the microchip's commands.

Materials List

Component	Price
General - Wheels - Screws - Bolts - Wood Panels - Nuts - Ribbon Cables	General - $5.03 RobotShop - $4.91 from Amazon - 4 x $6.19 from Home Depot - Provided - 2 x $8.19 from Amazon - 2 x $4.00 Spool3D
Line Detection - 1k Ω Resistor (R2) - 85k Ω Resistor (R3) - Superbright Led (D2) - Vishay BPW17N Phototransistor (Q3)	Line Detection - $0.15 from DigiKey - $0.37 from DigiKey - $0.48 from DigiKey - $1.29 from DigiKey
Wall Detection - 4 µF Capacitor (C5) - TPSA476K006R0800 47 µF Capacitor (C6) - Parallax 28015 Distance Sensor	Wall Detection - $0.43 from DigiKey - $1.15 from Mouser - 2 x $34.95 from Parallax
Flame Detection - 103k Ω Resistor (R4) - QSD123 Infrared Phototransistor (Q2)	Flame Detection - $0.87 from DigiKey - $1.08 from DigiKey
Fire Distinguisher Board - 1N4001G Diode (D1) - 1000 µF Capacitor (C1) - Fairchild Semiconductor TIP120 (Q1) - 10k Ω Resistor (R5)	Fire Distinguisher Board - $0.22 from DigiKey - $2.44 from Mouser - $1.72 from DigiKey - $0.14 from DigiKey
Motherboard - 100k Ω Potentiometer (R1) - NHD-0216K1Z-NSW-FBW-L LCD Display - 233k Ω Resistor (R6) - ATMEGA328P-PU Microchip	Motherboard - $1.88 from DigiKey - $15.14 from DigiKey - $0.24 from DigiKey

Motor Driver - TI SN754410NE Motor Driver Chip - 10,000 µF Capacitor (C3) - 33k Ω Resistor (R7 & R8) - Nema 17 Stepper Motor - Motor Bracket	Motor Driver - $5.74 from DigiKey - $13.52 from Mouser - 2 x $0.14 from DigiKey - 2 x $13.50 from Amazon - 2 x $5.99 from Tinkersphere
Total Price	$215.10

Timeline

February and March - Breadboarding

23 Working on proposal	24 Working on proposal	25 Working on flash LED	26 Troubleshooting flash LED	27 Working on motors	28 Weekend	01 Weekend
02 Testing directional logic	03 Working on line detection	04 Calibrating IR sensor sensitivity	05 Mapping sensor inputs	06 Trouble-shooting false positives	07 Weekend	08 Weekend
09 More trouble-shooting	10 Working on LCD	11 Wiring parallel interface pins	12 Programming and testing	13 Testing real-time status updates	14 Weekend	15 Weekend
16 March break	17 March break	18 March break	19 March break	20 March break	21 March break	22 March break
23 Working on wall detection	24 Testing avoidance logic	25 Working on flame detection	26 Testing IR flame sensor range	27 Trouble-shooting false positives	28 Weekend	29 Weekend
30 Working on distinguisher (fan)	31 Integrating and testing fan logic					

April - PCB Board

30	31	01 Working on motors	02 Soldering screw terminals	03 Mounting motor driver to chassis	04 Weekend	05 Weekend
06 Working on line detection (etching)	07 Creating custom headers	08 Ensure sensors sit at a fixed height	09 Cleaning up cable management	10 Testing and trouble-shooting	11 Weekend	12 Weekend
13 Finalize testing with actual lines	14 Working on LCD	15 Soldering pins	16 Mounting the LCD	17 Adjusting contrast with potentiometer	18 Weekend	19 Weekend
20 Testing LCD outputs	21 Working on wall detection	22 Ensuring sensors are level	23 Soldering capacitor to the power rail	24 Finalizing LCD (troubleshoot)	25 Weekend	26 Weekend
27 Working on flame detection	28 Soldering IR sensor	29 Finalizing sensor’s position	30 Testing and trouble-shooting			

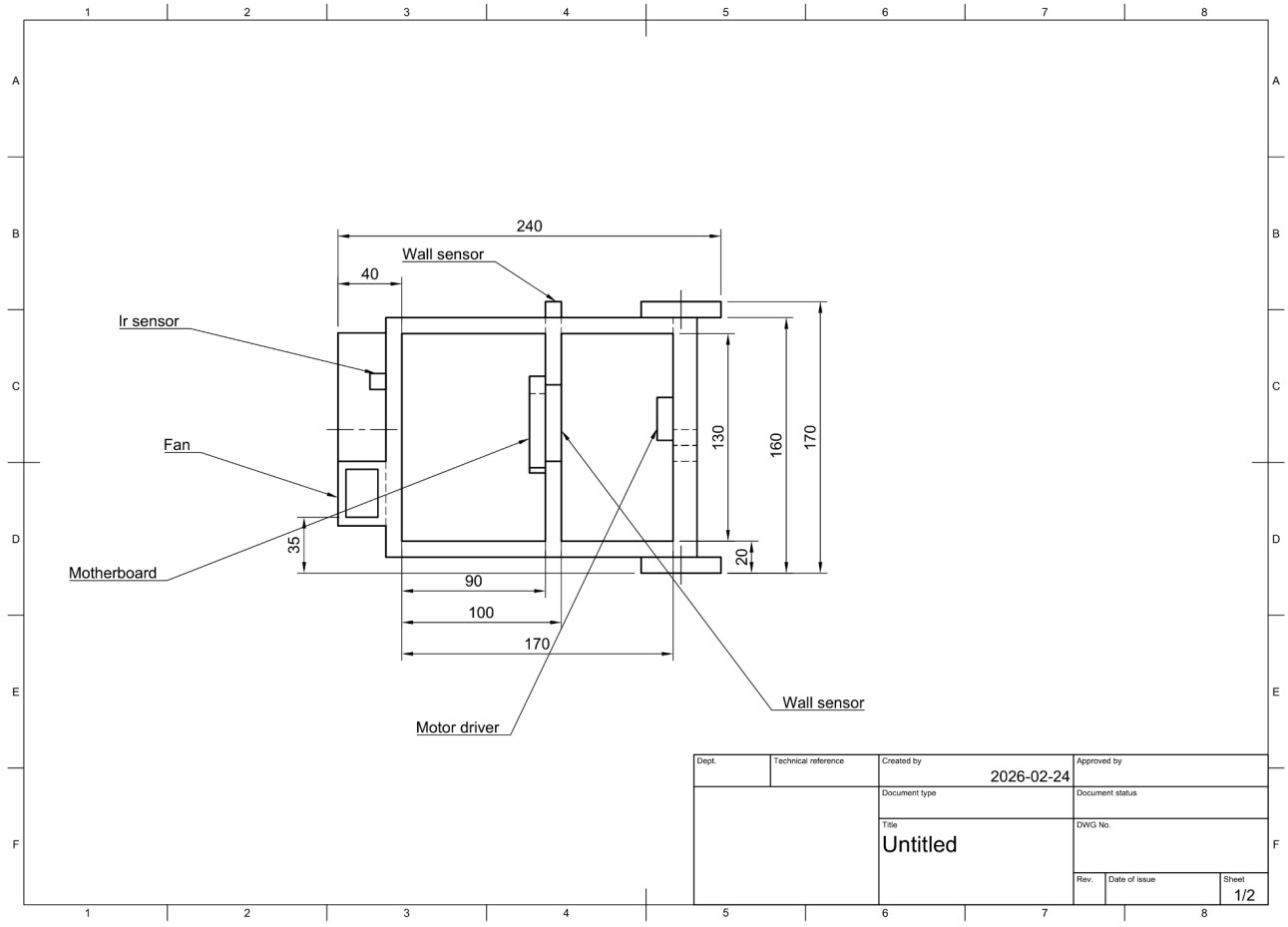
May - Finalizing Firefighter Bot

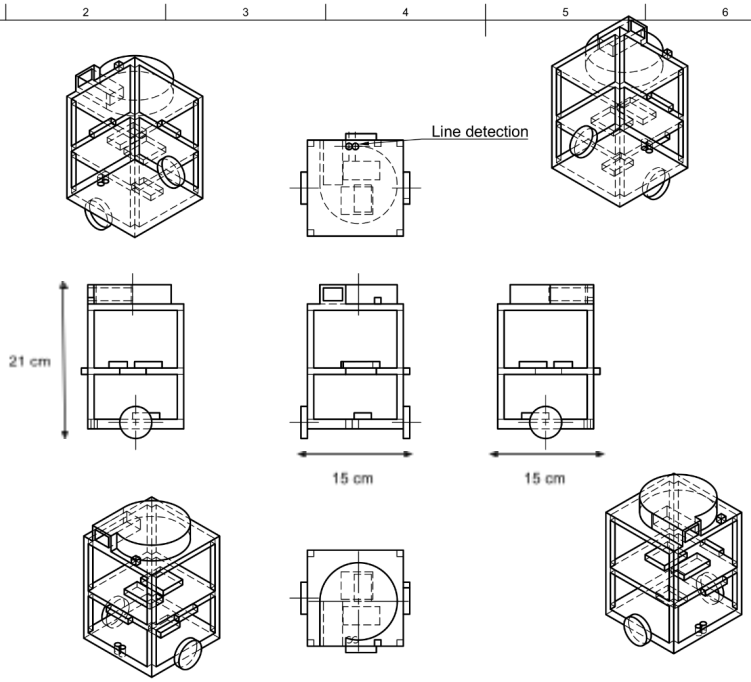
27	28	29	30	01 Working on distinguishing board	02 Weekend	03 Weekend
04 Soldering fan-driver circuit	05 Optimally securing fan to chassis	06 Trouble-shooting and testing bot	07 Trouble-shooting and testing bot	08 Trouble-shooting and testing bot	09 Weekend	10 Weekend
11 Trouble-shooting and testing bot	12 Trouble-shooting and testing bot	13 Trouble-shooting and testing bot	14 Trouble-shooting and testing bot	15 Trouble-shooting and testing bot	16 Weekend	17 Weekend
18 Working on bot in maze	19 Working on bot in maze	20 Working on bot in maze	21 Working on bot in maze	22 Working on bot in maze	23 Weekend	24 Weekend
25 Working on bot in maze	26 Working on bot in maze	27 Working on bot in maze	28 Working on bot in maze	29 Working on bot in maze	30 Weekend	31 Weekend

June - Completing the Mazes

01 Working on bot in maze	02 Working on bot in maze	03 Working on bot in maze	04 Working on bot in maze	05 Working on bot in maze	06 Weekend	07 Weekend
08 Working on bot in maze	09 Working on bot in maze	10 Working on bot in maze	11 Working on bot in maze	12 Working on bot in maze	13 Weekend	14 Weekend
15 Working on bot in maze	16 Working on bot in maze	17 Working on bot in maze	18 Working on bot in maze	19 Working on bot in maze	20 Weekend	21 Weekend
22 Working on bot in maze	23 Working on bot in maze	24 Working on bot in maze	25 Working on bot in maze	26 Summer	27 Weekend	28 Weekend
29 Summer	30 Summer	31 Summer				

Drawings


240
Wall sensor
40
Ir sensor
Fan
Motherboard
35
90
100
170
Motor driver
Wall sensor
130
160
170
20
Dept.
Technical reference
Created by
2026-02-24
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Document type
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